AI in Healthcare

Submitted To: Mr. Chintan Patel

Submitted by: Devanshu Joshi

Enrolment Number: 20261015

Group 03

Executive Summary:

A large hospital chain in India can benefit from using intelligence (AI) to reduce mistakes in diagnosis improve the care given to patients and use resources like beds, medical staff and diagnostic facilities more efficiently. However just making predictions is not enough to make healthcare services better. For example an AI system can spot a patient who needs help but the situation might not get better if lab reports are late specialists are not there beds are not available or no one is responsible for taking the next step.

This report introduces the Clinical–Operational Care Command (COCC) a system that helps make decisions under the guidance of doctors and nurses. It wants to connect alerts about risks with the actions needed in the clinic and the resources available. The system would bring together information from health records monitoring of vital signs, lab and imaging reports, medicine records, bed management systems and how patients are followed up after their visit. It would add to existing hospital systems of replacing them. It would not make diagnoses on its own prescribe medicine approve leaving the hospital or decide how to use medical resources.

The report suggests starting a test project in the emergency room focusing on patients who might have sepsis or signs that their health is getting worse. Studies show that systems that warn about sepsis might help with following treatment and making patients feel better; however the results are different in places. So the results from studies can't be used directly for the hospital chain in question (Kim et al. 2024). Also an outside check of a sepsis prediction model found that it might not work well as expected in a new healthcare setting (Wong et al., 2021).

Introduction and Case Context:

Hospitals create a lot of data from patient records monitoring of vital signs, lab tests, radiology services, pharmacy systems and tools for managing beds. Artificial intelligence can help doctors and nurses choose the information spot possible dangers and guess what resources will be needed in the future. However these abilities can work on their own in hospital systems. Because of this an alert might be. Not reach the right person in time and the teams that manage operations might not know which delays need attention right away.

The problem in this case has two parts that're very close to each other: the chance of making mistakes in diagnosis and not noticing when a patients condition is getting worse which can slow down the needed tests and treatment; and the fact that the ways of working are not connected and there are not enough resources that can stop action from happening even after a patients risk is known. Overcrowding in the emergency room has been linked to delays in care for sepsis; however this link does not show that adding an AI system will lower the number of deaths.

The main issue is the space between knowing there is a risk and finishing the care that is needed. The solution aims to close this space by using AI help for decisions along with responsibilities steps to take if things get worse and checking how well things are done. The report focuses on a made-up hospital chain in India. It does not say that any real hospital has already used the COCC system.

In a hospital chain a patient may go through many departments and digital systems during one visit to the hospital. People who check patients first write down their symptoms and vital signs doctors ask for tests and scans specialists look at the results nurses give treatment and the teams that manage the hospital coordinate where people sleep, move and leave. When people change shifts important information can be delayed, not noticed or not followed up properly. AI can help bring the right information; however its help depends on whether there are clear steps to take based on the information given.

So the goal is not just to put a model that predicts things into the hospitals work. Instead the focus is on making the care process better starting with finding a possible risk and going all the way through talking about it giving responsibility fixing problems with the work and writing down what was done. An alert might not be very useful if no one is responsible for looking at it. In the way predicting how many beds are needed might not help move patients better unless it is connected with staff plans and when people leave the hospital.

This report looks at a made-up hospital chain in India of looking at a real hospital. For the case study the hospital chain is assumed to use a mix of health records, lab systems, radiology systems, pharmacy databases and tools for managing beds. The real abilities and how well these systems are connected would need to be

checked before putting the COCC in place. The COCC would be a system that connects existing ones as much as possible while still letting doctors and nurses do their jobs and not replacing the old records.

AI Applications Across the Patient Journey:

Early Disease Detection and Clinical Risk:

Machine-learning models can look at things like signs, lab results, medical history and changes in a patients condition to find possible health risks earlier than might be seen by just looking at each part alone. For people who might have sepsis an AI risk score could tell the team collect the important test results and say when more checks might be needed. However the score should be seen as a sign of danger, not a confirmed diagnosis.

A review and study from 2024, which looked at 22 studies with 19,580 patients in emergency rooms found that systems that warn about sepsis were linked to a chance of dying, shorter stays in the hospital and better following the rules for treating sepsis. However the people who did the study also saw differences between the studies, which means the results should be looked at carefully (Kim et al. 2024). These results are about systems for sepsis alerts and don't prove that all AI models or the COCC will have the same results.

Wrong alerts are another issue because they can cause extra tests, wrong use of antibiotics getting tired of alerts and less trust in the system. A report on checking how well the Epic Sepsis Model worked in 38,455 hospital stays found an area under the curve (AUC) of 0.63. At the threshold being checked the chance of finding the problem was 33%. The chance that the alert was right was only 12% (Wong et al. 2021). These results show why hospitals need to check how well the model works and fits with their patients and how they do their work before relying on it.

Personalised Treatment and Clinical Decision Support:

Artificial intelligence can help doctors and nurses give custom care by looking at a patients medical history pointing out possible problems with medicine finding important guides and helping plan what happens next based on the patients risks. These abilities might help doctors think about information more often especially for people who have many health issues. Of giving unclear or unexplained care plans the system should show the right information, the reasons for it and any uncertainty with the suggestions.

The COCC system would not give medicine on its own decide on the treatment plan or let patients leave the hospital. These jobs would stay with trained doctors and nurses who would be in charge of explaining their choices to patients. Whenever the system gives a suggestion it should keep a record of the information that was used to make it. Doctors and nurses should also be able to say no to a suggestion and explain why during safety reviews

Diagnostic Worklist Prioritisation:

Using AI to decide what to do first can help hospitals find findings in scans and lab tests so they can get attention quicker. This method works best where the specific problem, the way it is found and how the work is done has been tested and checked in the hospital. Even if the time it takes to report something goes down that doesn't mean patients are getting better. The job of understanding the results and deciding what they mean still belongs to the doctor or the right healthcare professional.

With the COCC way of doing things AI should first be used for urgent cases. The system should keep a record of when a finding was marked, who looked at it and how long it took to respond. Hospitals should check the time saved. Also think about possible risks like putting less urgent cases first and delaying other patients who need help.

Predictive Capacity and Hospital Operations:

Predictive models can guess how many people will come to the emergency room how many beds will be used how many people will need care how much work there will be in labs and when people will leave the hospital. This helps the teams that manage the hospital plan staff schedules find problems and fix them before they become real issues. However predictions can be uncertain especially when the people how staff there are, the time of year and the rules in the hospital change.

The COCC system should use these predictions to help plan and handle problems not to make choices about who gets a bed who is turned away or who comes first. When deciding which patient needs help first the need for care should always be the top priority. Information on what resourcesre available should help doctors and nurses plan the right response without putting patients in danger.

Even though these four parts of using AI are connected by the patients journey each should be looked at on its own. Early warning systems, tools for finding cases help for treatment and models for predicting how many people will need care all serve different roles, with different people, limits and safety issues. Looking at them with one measure of how well they work could hide important differences. Each part should therefore have a medical goal, a group of patients it aims at, a way to act steps to take if things go wrong and a way to keep checking how well it is working.

Follow-up, Discharge Planning and Continuity of Care:

AI can help plan when patients leave the hospital by helping the care team find tests that are not done appointments that are not made medicine reviews that are not finished and things that need to be arranged after leaving. Predictive models might also help find patients who might need help after leaving. However these predictions need to be balanced against the patients health, their situation and what is happening in their home. The system should push the care team to think about the situation not just say a patient is not ready to leave or make the final choice to let them go.

One practical use would be a checklist that shows what is still needed for a patient to be discharged like test results that are not in yet instructions for medicine that are not clear or a follow-up appointment that has not been made. The system needs to tell the difference between missing information and something that is actually needed for the patient to be safe – just because a box is empty does not mean there is a problem. Staff should have the option to say a task does not apply and add an explanation if needed. The success of the tool can be checked by seeing if important parts of the discharge are done, if there are delays that could have been avoided if the patient understands what is happening and if follow-up happens, not by looking at how often the patient comes back to the hospital.

Generative AI for Clinical Documentation:

Generative AI can help doctors and nurses by summarizing patient records making draft notes from approved conversations and getting the right hospital rules. These tools can save time searching and making notes. The information made by the AI can miss important details add things that are not true and mix up what the patient is going through now with what happened before. To cut these risks the system should link the things it says back to where they came from show what is made by the AI and make sure a doctor checks the information before it goes into the official record.

The hospital should not let people put patient details into a public chatbot unless the service has been checked and approved under the hospital's rules for privacy, security and getting things. Before starting any test of the documentation tools the hospital needs to know what information can be used, where it will be kept how long it will stay and who can see it. The check should also look at how much editing's needed and how much time is saved because a draft that needs a lot of fixing does not help the work much.

Proposed Clinical–Operational Care Command:

The Clinical–Operational Care Command (COCC) is an AI tool that is meant to help with the patient's journey in the hospital. It is meant to link signs of problems with the right steps to take the tests that need to be done the things the hospital has and the tasks that are done. It will work with the existing hospital systems not replace them or make decisions that need a doctor's judgment.

A key part of the plan is that the hospital's resources should never affect how important a patient's health problem is. If a patient who is at risk needs something that is not available COCC should show that there is a problem and tell the right people. It should not change the patient's priority just because the hospital does not have what is needed now.

For the test the system may need the patient's name, basic details past medical history signs like temperature and blood pressure lab tests and results some medicine records, important times test queues, where the patient is, how many beds and ICU spots are available and what the staff did for the tasks. Only what is needed for

the test should be used. Getting the times right and making sure the patient is identified properly is very important because mistakes here can lead to predictions, which makes the test results not useful.

The main goal of COCC is to connect a warning from the clinic to a next step. Of putting everything into a hard-to-read score the system should show why the warning was given, how urgent it is, the proof behind it if there is any doubt what needs to be checked and if the task is done or delayed. Medical staff should be able to see if a warning is waiting to be looked at has been accepted or changed 's done or has been put off.

The main idea is that medical urgency decides the patient’s place in line while the operational info helps the hospital arrange and do what is needed. If COCC sees a patient who's in a lot of danger but there is no bed available the system should tell the doctor and the right person in charge but it should not lower the patient’s status because there is no bed. Any way of sorting patients by how urgent they're should be approved by the hospital’s medical leaders, tested in real situations and used by the right people. The hospital should not decide who is more important based on if a service's available.

The data collected should only be what is needed for the test. This might include the times when things were checked the history that matters, lab tests and results what the doctors looked at, where the patient is and some times related to the work. The hospital should say where each part of the data comes from how often it changes, what to do if it is missing or not matching and what to use if there is a conflict. Duplicate entries late updates and wrong times can make the model less correct even if the algorithm is right.

The team working on the system should test it with problems before using it in a real hospital. Possible problems include lab results that come late moving a patient between areas starting treatment before a warning is given the system not working and information that is different on platforms. The screen should show when the info is old and the software should have ways to use it safely when the model is not working. If the AI or the data is not working the hospital should use the ways of doing things.

Medical leaders should say what the tool is for, how to know when to warn what to do if the warning is not answered and when the test should end. Doctors and others should always make their choices about each patient. Nurses and staff can pass on warnings if they are allowed to. People in charge of beds and moving people should handle those parts. The people in charge of the computer part should make sure the data is right and the system works. A group should check on any problems what doctors change how the model is. Any changes that need to be made.

Warnings should be given to people and have a time when they should be answered, not sent to many people without saying who has to do it. There should be steps to follow if a warning is not answered, with the time it took to respond recorded and reviewed for better ways. Doctors can say a warning is not needed. They should say why so that it can be checked if it was right or if there is something to learn.

Pilot Implementation: Sepsis:

The test should start at one hospital or in the emergency room where the doctors are interested. The team should look at the steps from when the patient comes in through the first check seeing the doctor, tests, plans for treatment and where the patient goes. They should check what data is there the times, who's in charge and how to handle the problems. They should set a baseline to know how things are before the AI starts to help.

The model should start in a way where it makes predictions but does not send out warnings. Its results should be checked against what the doctors see and what is the usual way of doing things. The team should look at how good the model's how likely it is to be right how often it is correct how well it works for different groups of people and how early it can warn. They should guess how warnings there are each shift and if the answers are good. The test should not move to the step if there are big safety problems, data or work issues or too many warnings.

When the special part is done the hospital can think about turning on a warnings with doctors looking at them. This is also a time to train doctors, nurses, lab workers, radiologists and people who manage beds. The staff should know how to say when a warning is confusing when there is a problem with the system or when something is not safe. The test should check if warnings are changed how quickly they are answered, if there are warnings and any other problems at least once a week.

These results should be compared with the starting point. If possible, with other places that do not use this or use it later. The check should take into account differences in patients the way the work is done the people working and other things. A small test might show that the idea works and helps some things without proving that it saves lives because the number of people is too small. The hospital should think about a test if it meets the rules about safety how the staff reacts, the data quality, fairness and how it affects work.

Sepsis is a condition to use for the test because it has many signs and needs quick steps. At the time it is hard to diagnose because the signs are not clear the data is not complete and an early warning might not be right. COCC should help not replace what is already there. In particular the way a patient is treated should not change just because the model says something.

The hospital needs to decide what the model is trying to predict and how far in advance the warnings should be. For example predicting that a patient will get worse and asking a doctor to check at a time are two different goals. The test should set rules for who's included and who is not and note if some groups are missing data. The people should know what to do if the model is not working or if the data needed is not there

During the part a group of trained people should look at what the model says and compare it with what happens to the patient. The check should look at both bad cases to find ways to get better. The people checking should ask if a warning would have made a difference and if the information was there to do something about it. The model’s results should be checked on groups of patients to make sure it does not need different care for some people. Even though it helps the special part is not the same, as a test, which is needed to see how doctors really respond to the warnings.

The training for the pilot should be practical and relevant to the staffs responsibilities. They need to understand the context and limitations of the information provided by the model record their responses appropriately and follow established procedures for the models failure. The staff should report problems, duplicate alerts, missing or incorrect information and inappropriate response times for review. The implementation team should consider these insights and operational concerns when making adjustments during the first weeks of the pilot

The decision to adopt the model should not be based on enthusiasm or extrapolation from one hospitals experience. The hospital should halt the process if it discovers cases inappropriate alert load, significant differences in performance between different populations, unacceptable operational burden or unresolved privacy or security concerns. Improved response times without considering the opportunities for investigations or treatment would be inappropriate. The hospital should only consider expansion if there is a safety and operational benefit.

Implementation and Governance in the Context:

A hospital group may have different electronic health record systems, laboratory data platforms, picture archiving servers and bed management systems at different hospitals. The technical infrastructure should be able to assign patient identifiers use common terminologies, record timestamps accurately utilise appropriate access controls and have necessary interfaces. The Ayushman Bharat Digital Mission develops standards for consent-based health information exchange. The hospital group should review the requirements and specifications to its intended use and determine whether the technical infrastructure meets the applicable standards. Not all digital patient record management systems are inherently compliant with the framework.

The hospital should review the requirements of the Digital Personal Data Protection Act, 2023 and the Digital Personal Data Protection Rules, 2025 including those that may be in force at dates. The implementation plan should address processing, notice and consent requirements, access controls, retention, breach management, vendor terms and patient grievance resolution mechanisms. The legal team should ensure that only the requirements that are directly applicable to the given pilot are considered than assuming that all provisions are in force.Medical device laws may apply to AI models. On 21 July 2026 the Central Drugs Standard Control Organisation (CDSCO) issued a Guidance Document on Medical Device Software under the Medical Devices Rules, 2017. The hospital and its vendors should determine whether each clinical AI component falls under the framework what its classification is, what documentation, quality management system requirements and post-market surveillance activities are applicable. Operational forecasting applications and those with clinical recommendations may have different regulatory requirements and each component should be reviewed accordingly.

The World Health Organizations guidance highlights the importance of ethics, human rights, accountability and responsiveness in the development and use of AI (WHO, 2021). They should be embedded in the principles of the COCC system, which requires a defined intention, local data validation, clinician oversight, documented responsibility, evaluation for biases and a process for model suspension when appropriate.

Fairness considerations should include differences in distributions, false-positive rates time spent reviewing alerts and task completion across different groups of patients and hospitals. The language and practices should be appropriate to the locations where the model is used. Security practices should include access control, encryption, auditing, vendor access limitations, regular backups and procedures for continued operations during disruptions. Patients and staff should be able to raise concerns and file complaints.

Interoperability should be considered a safety requirement because it helps clinical staff receive the information to make timely decisions. Consistent patient identifiers, temporal information standardised clinical terminologies and reliable data exchange between different applications are essential to ensure appropriate clinician responses. Incomplete delayed or incorrect data or assignment to the patient will lead to incorrect alerts and unnecessary interventions. Connectivity and responsibility for data integrity need to be established before relying on COCC for operational decision-making.

The system should ensure that each alert is associated with the patient and episode of care. Before applying approaches across the hospital group the variability between hospitals should be understood. An approach that has been trialled in one hospital may not produce the results elsewhere due to differences in populations, staffing, documentation and diagnostic tests available.

Interoperability initiatives underpinning Ayushman Bharat Digital Mission could provide guidance on consent for sharing information and standardised recording of health data. However a hospitals participation in an information network does not automatically ensure that each application of intelligence (AI) is lawful, confidential or fit for purpose. The hospital should agree a set of standards and internal control policies that apply to its information and activities. It should also establish a record of rationale for accessing information and avoid retaining information beyond the purpose of care.

Confidentiality and security could be achieved by ensuring that access is restricted according to roles data is encrypted when stored and transmitted audit trails are in place information is retained for as long as necessary there are regular backups and recovery tests response to incidents is appropriately managed and third-party access is limited. Using pseudonymised or de-identified information for model training could reduce privacy risks. Should not be viewed as an exemption from other governance requirements or a guarantee that individual patients could not be re-identified. Contracts with suppliers should address ownership of information and models updates and sub-processing, disclosure of incidents, continuity of service and access to information for regulatory inspections, including the right to delete information.

Indias data protection legislation includes the Digital Personal Data Protection Act, 2023 and the Digital Personal Data Protection Rules, 2025. Before implementation the hospital should seek advice on its obligations, exemptions and responsibilities for each category of processing. It should not assume that everything applies from a date stated in the legislation; each obligation may come into force at a different time. Software. Modified for clinical purposes may also be subject to medical-device regulations depending on its intended purpose and the jurisdiction. The regulatory status of each product should be clarified before purchasing or adopting it.

Before implementation the hospital should. Document a risk assessment of the proposed model. Technical and clinical risks should be weighed up against the benefits taking into account the severity of the harm, likelihood, groups of patients who may be affected the possibility of detecting errors and the availability of approaches if the model fails. Significant changes such as alert thresholds, additional populations or a change in supplier should also undergo review and re-validation. The hospital should not blindly adopt model updates proposed by the vendor.

Continuous Monitoring:

The effectiveness of the model should be evaluated using a operational and safety assurance framework instead of relying on a single metric such as overall accuracy. The hospital should set out reference values or

targets for each indicator based on conditions at the time of implementation and consider improvement as an iterative process. It should not assume that changes are beneficial unless they have been evaluated.

Clinical effectiveness was measured in terms of mortality and length of hospital stay or readmission. In each case requires sufficient numbers of patients careful adjustment for patient risk factors and a valid comparison method. A change in one of these outcomes before and after implementation of COCC does not prove causality. The evaluation should consider non-safety related benefits such as inappropriate allocation of resources, unnecessary investigations or treatments and delays to care for those not receiving an alert.

Monitoring should start with establishing a baseline representing the hospitals performance before the intervention. Each indicator should have a definition, source of information inclusion criteria where relevant and method of managing missing data. Wherever possible and lawful the hospital should analyse subgroups such as hospitals, wards, age or sex. Interpret findings with caution if a subgroup analysis has a small sample size that could influence the results or disclose personal information.

The monitoring of performance and impact on operations should be performed separately. Statistical metrics such as area under the receiver operating curve should be interpreted with caution as standalone indicators of clinical effectiveness, appropriateness of calibration or safety at a given threshold. The evaluation should also consider sensitivity, specificity, positive predictive value, calibration, positive and false-negative rates and these should be weighed up against the intended use of the model and prevalence of the outcome in the local population. It is equally important to assess how clinicians responded to the information provided by the model.

Operational performance can be measured by the time from alert to review from test ordered to result available specialist review, escalation task completion and the number of tasks at any one time exceeding a predefined threshold. The hospital should also consider indicators addressing sources of underuse such as the number of alerts raised per shift unnecessary investigations or treatments initiated extra staff time spent responding to the model delays to other patients caused by the intervention and the number of clinician overrides. Term clinical indicators such as mortality, length of stay and readmission can be useful to evaluate overall impact but require careful interpretation due to small sample sizes the need to adjust for differences between patients and potential interactions with other changes in healthcare provision.

The final evaluation report should include inconclusive findings alongside positive changes. If a specific indicator changed after the implementation of AI the hospital should not attribute this to the technology unless there is evidence. A difference in mortality length of stay or readmission following the introduction of COCC could reflect differences in staffing or patient populations well as seasonal variations, changes in treatment pathways and other initiatives undertaken at the same time. Wherever possible the hospital should use methods of evaluation such, as a control group or staggered roll-out to measure real-world benefit. The main aim of evaluation should be to demonstrate value compared to the prevailing standard of care than prove that AI works.

Risks, Safeguards and Recommendations:

The hospital needs to use an specific suspected-sepsis pathway try it in shadow mode and only adopt features that have a clear purpose, a responsible person and a proven clinical benefit. It should not put a lot of effort into a platform just because it might help with integration without first defining clinical questions and checking results. Another important safety measure is making sure that doctors understand the limits of the technology and do not let it replace their skills. The hospital should train staff to see an AI alert as a signal to check the patient again of just confirming a diagnosis. It should also make sure that doctors are not flooded with repeated or unhelpful alerts about the problem. The rules for when alertsre sent and when duplicates are hidden should be looked at using local data. At the time the hospital must keep the ability to get to the basic information and stop life-saving alerts from being ignored automatically.

Not having good quality data can cause problems in spotting when a patient is getting worse or causing increases in care. For example if a patient is getting fast the lack of a measurement should not be seen as normal. Also old observations need to be marked so they are not used by mistake for decisions. The system should show where each important number came from and when it was recorded. If the information is missing or the patient does not fit the models rules the system should tell doctors to follow the care process.

Using AI to make things more efficient by managing beds, staff or equipment might conflict with the hospitals duty to give care based on patients real needs. If the system predicts not space the hospital should take action to handle the demand and increase the supply instead of turning away patients. Also if the system suggests delaying a test because the list is full it should not stop doctors from doing the test if a patient needs it away. The hospitals clinical governance committee should decide how to handle these issues and make sure staff can talk about concerns when AI alerts are used without rules.

The hospital should think about using a step-by-step way to introduce the system and do a study before using it widely. It should follow the care process find an area to improve that can be measured check the data do a shadow test and then run a small supervised implementation. It should also create a place for all AI tools in the hospital group explaining what they are for who's in charge who made them the version of the model how they were checked and any problems that happened to help with checking and being responsible.

Artificial intelligence can help hospitals spot patients who're in danger give more personal care decide which tests to do first and predict how much resources are needed.. The good things from these tools depend on the situation they are used in and how well the predictions lead to quick actions by doctors. The suggested Clinical–Operational Care Command (COCC) is meant to help hospitals close the gap between finding a problem and solving it by connecting risk alerts to the care tasks showing available space the right way to get help and checking things regularly. A sepsis test could be a way to try this idea. At the time the hospital should make sure that the good things from these systems are not assumed to be the same everywhere but are checked locally and that things like staff and how the hospital works are considered along with the technology. In general the hospital should be careful when using AI tools and only expand them if they have been shown to make care safer and more efficient without hurting care.

Overall the hospital should see AI as a way to help doctors provide efficient care. Early warning scores, choosing which tests to do advice on treatment help with writing notes and predicting how much space is needed can all help reach that goal. But just having a model available does not mean it works and each tool needs a reason for being used checks on the data and local control to make sure it is right for the job and used safely in the current ways of working.

The suggested COCC system is an idea to handle the problem in a made-up hospital chain, which is the difference between spotting clinical risks and actually doing the needed treatments. A test on finding sepsis cases could help see if it really reduces the time doctors spend on checking patients without making the staff more tired. At the time it is important to know that this system might have bad effects like possibly being unfair, to certain patients. However these good points are only possible if the hospital chain can show that the system works with its data. So the right thing to do in this case is to stop the test if the evidence does not support the idea.

Conclusion:

AI can help us in detecting risks early, offering individualised decision support, prioritising diagnostics, and planning hospital capacities. Its practical value depends on the outputs being trustworthy for the local population, and connected into a workflow which allows for timely human action. COCC addresses the gap by connecting clinical signals to care tasks, to visible operational constraints, to responsible escalation, and to closed-loop tracking.

A suspected-sepsis pilot offers a focused entry point to test the concept, while expected benefits remain hypotheses. Local validation, human oversight, careful governance, and balanced evaluation are required before benefits can be claimed. The strongest recommendation is a controlled and evidence-based roll-out, which only grows if safety and meaningful value can be demonstrated.

The central recommendation is seeing AI as a way to strengthen the coordination between clinical insight and reliable execution. Early-warning models, diagnostic prioritisation, treatment support, documentation support, and capacity forecasts can all contribute, but none should be expected to generate benefits just by existing in technical capability. Each requires a well-defined task, appropriate data, local validation, a safe production workflow, and an accountable human decision-maker.

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Appendices: Supporting Tables

Appendix A: COCC Responsibilities and Human Oversight Table

This table provides an overview of how the proposed system can be used to support coordination while ensuring clinical judgment and accountability by specific staff.

COCC may support	**Humans retain authority**
Risk flags and review-task generation	Diagnosis and clinical interpretation
Visibility of diagnostic queues and delays	Treatment and medication decisions
Demand and capacity forecasts	Admission, discharge, ICU transfer and scarce-resource allocation
Follow-up reminders and unresolved-task tracking	Patient communication, consent and final closure decisions
Audit dashboards and performance monitoring	Approval of model changes and safety policy

Appendix B: Evaluation Measures Table

The following measures are recommended for local evaluation. The baseline and targets should be determined, and no improvements are expected.

Dimension	**Measures**	**Interpretation**
Clinical safety	Missed/delayed critical findings; false-negative events; alert PPV; calibration; safety incidents	Detects unsafe predictions and downstream harms.
Timeliness	Time from flag to review, test completion, specialist review and treatment decision	Shows whether workflow becomes faster; does not alone prove better outcomes.
Appropriate care	Guideline-concordant assessment; unnecessary testing or antibiotic use; clinician overrides	Balances speed with clinical appropriateness.
Operations	Diagnostic turnaround; unresolved queue time; bed/ICU constraint duration; escalation resolution time	Shows whether bottlenecks are identified and resolved.
Adoption/workload	Alerts per shift; acknowledgement; override reasons; staff feedback; task closure	Indicates workflow fit and alert-fatigue risk.
Equity/privacy	Performance by subgroup/site; differential waits; privacy incidents; access-log exceptions	Identifies unequal impact and governance failures.

Appendix C: Key Risks and Safeguards Table

This table contains key implementation hazards and safeguards that should be considered when designing and implementing the pilot.

Risk	**Safeguard**
False positives and alert fatigue	Shadow-mode testing, tiered alerts, locally tuned thresholds, clinician feedback and weekly review.
Poor data quality or missing timestamps	Data-quality dashboard, reconciliation checks and minimum readiness criteria.
Model drift or subgroup bias	Ongoing calibration and subgroup monitoring; revalidation after major changes.
Operational bottlenecks remain unresolved	Named escalation owners, response-time expectations and leadership review of repeated constraints.
Overreliance on automated output	Human decision authority, visible uncertainty, override capability and training.
Privacy, cybersecurity or regulatory gaps	Privacy and security review, access controls, vendor assessment and component-specific regulatory review.